61. The mean yield of rice observed from randomly selected 100 plots in District A was 210 kg *per acre* with standard deviation of 10 kg *per acre*. The mean yield of rice observed from randomly selected 150 plots in District B was 220 kg *per acre* with standard deviation of 12 kg *per acre*. Assuming that the standard deviation of yield in the entire state was 11kg, test at 1% level of significance whether difference between the mean yields of rice in the two districts is significant.

62. The standard deviation of the scores secured by XI standard students in a test conducted for examining their numerical ability is known to be 15. A school implemented a new method of teaching which is supposed to increase general quantitative ability. A group of 99 students are randomly assigned to one of two classes. Fifty students in Class-I are given the new method of teaching, whereas the remaining 49 students in Class-II are taught in the standard way. At the end of the particular term, students are given the same test of ability. The average scores secured by the students studied in Class-I and Class-II are respectively 116.0 and 113.1. Does this information provide significant evidence at 5% level, to conclude that the new method improved the numerical ability of students?

63. A machine assesses the life of a ball point pen, by measuring the length of a continuous line drawn using the pen. A random sample of 80 pens of Brand A have a total writing length of 96.84 km. Random sample of 75 pens of Brand B have a total writing length of 93.75 km. Assuming that the standard deviation of the writing length of a single pen is 0.15 km for both brands, can the consumer decide to choose Brand B pens assuming that their average writing length is more than that of Brand A pens? Set level of significance as 1%.

64. A study was conducted to compare the performance of athletes of two States in Inter-State Athlete Meets. Details of the number of successes achieved by the athletes of the two States are given hereunder:

State	No. of Athletes	Mean	Standard Deviation
State-1	300	75	10
State-2	400	73	11

Does the above information ensure at 1% level of significance that the difference between the performances of the athletes of the two States is significant?

65. A District Administration conducted awareness campaign on a contagious disease utilizing the services of school students. Among 64 randomly selected households, 50 of them appreciated the involvement of students. Can the District Administration decide whether more than 90% success could be achieved in these kinds of programmes by involving the students? Fix the level of significance as 1%.

66. A coin is tossed 10, 000 times and head turned up 5,195 times. Test the hypothesis, at 5% level of significance, that the coin is unbiased.

67. A study was conducted among randomly selected families who are living in two locations of a district, and parents were asked "Whether watching TV programmes by parents affects the studies of their children?" Details are presented hereunder:

Locality	No. of Families	
	Contacted	Agreed
A	200	48
B	600	96

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